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Optimization for cooperative task planning of heterogeneous multi-robot systems in an order picking warehouse

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ABSTRACT

This article investigates the cooperative task planning problem of heterogeneous multi-robot systems in an order picking warehouse, with challenging characteristics such as heterogeneous agents, interrelated utilities and complex intertask dependencies. These characteristics bring great challenges for modelling and optimization of the problem. This study establishes a novel mapping mechanism to describe the problem as open shop scheduling with sequence-dependent set-up and transportation times, and develops an efficient mixed-integer linear programming model to obtain the optimal task planning scheme for small-size problems. To increase the size of solvable problems and improve the performance of the solution, an effective hybrid artificial bee colony algorithm is proposed, which introduces some concepts from the tabu search algorithm and the artificial immune algorithm. The effectiveness of the proposed method is confirmed by simulation experiments.

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Cooperative task planning; heterogeneous multi-robot systems; order picking warehouse; hybrid artificial bee colony algorithm; open shop scheduling problem

1. Introduction

With the rapid development of economic globalization and technological progress, warehouse management plays a vital role in the global supply chain process (Yener and Yazgan 2019). As one of the core functions of warehouse management, order picking addresses the retrieval of items from their storage locations in response to customer orders (Masae, Glock, and Grosse 2020). It is recognized as the most time-consuming and labour-intensive operation (Wang, Zhang, and Fan 2020), and accounts for around 55–75% of total warehousing costs (Lu *et al.* 2016). Therefore, extensive studies on order picking have been conducted over the past few years (Ho and Lin 2017; Ardjmand *et al.* 2018; Lee, Chung, and Yoon 2020). Even though many attempts have been made to automate the picking process, systems involving manual operators are still prevalent in practice.

In general, there are two kinds of manual order picking system, namely picker-to-parts systems and parts-to-picker systems. In picker-to-parts systems, order pickers drive or walk through the warehouse and collect the required items, whereas automated storage and retrieval systems deliver the items to stationary order pickers in parts-to-picker systems (Lu *et al.* 2016). The former type is more common, accounting for more than 80% of all order picking systems (De Koster, Le-Duc, and Roodbergen 2007). However, in manual picker-to-parts systems, the time spent by pickers travelling in the warehouse accounts for approximately 50% of the total order picking time (Bottani, Volpi, and Montanari 2019). Therefore, a promising way to reduce the total order picking time would be by introducing fast-moving robots into the system to partly replace human functions for frequent long-distance movements (Li *et al.* 2020). One such order picking system is a typical heterogeneous

multi-robot system (MRS) where agents are distinguished by their specialized capabilities, *i.e.* object delivery capability and object pick-up capability. This study aims to improve the order picking operation by properly arranging for a group of cooperative robots to go to different locations to fetch specific items with the help of manipulators (Van Gils *et al.* 2018; Wang, Zhang, and Fan 2020).

In this article, the cooperative task planning of heterogeneous MRSs in an order picking warehouse is described as open shop scheduling with sequence-dependent set-up and transportation times (OSS-SSTT) through a novel mapping mechanism. Then, an efficient mixed-integer linear programming (MILP) model is developed for small-size problems. Owing to the NP-hard nature of the problem, the computational effort required to find an optimal solution grows exponentially with the problem size. To increase the size of solvable problems and improve the performance of the solution, an effective hybrid artificial bee colony (HABC) algorithm is proposed.

The remainder of this article is organized as follows. Section 2 provides a brief review of related studies on heterogeneous MRSs. Section 3 describes the cooperative task planning of heterogeneous MRSs and develops a MILP model for OSS-SSTT. Section 4 presents the HABC algorithm. In Section 5, experiments are conducted to investigate the effectiveness of the proposed algorithm. Finally, conclusions and future directions are summarized in Section 6.

2. Literature review

Research on multi-robot systems has been conducted since the late 1980s (Asama, Matsumoto, and Ishida 1989), to address some tasks that are intractable to achieve for a single robot with low cost and high system reliability (Yan, Jouandeau, and Cherif 2013). These benefits have driven many researchers from academia and industry to investigate the applicability of MRSs in commercial areas such as order fulfilment systems (Roy *et al.* 2019), search and rescue (Din *et al.* 2018), environment exploration (Wang, Wang, and Liu 2016) and unmanned aerial vehicles (Coutinho, Battarra, and Fliege 2018). According to the capabilities of the robots in MRSs, MRSs can be homogeneous or heterogeneous. However, when performing complex tasks, heterogeneous MRSs can achieve high adaptability at low cost without reducing fault tolerance compared with homogeneous MRSs, thus gaining wide attention (Chand and Carnegie 2013).

In heterogeneous MRSs, reasonable task planning is required to enable teams of agents to execute missions in a coordinated manner. These missions often entail completing discrete tasks at distinct locations to maximize reward while minimizing cost and adhering to spatial and temporal constraints (Gerkey and Mataric 2004). One focus of research is enhancing the performance of heterogeneous MRSs by improving the coordination technique (Cao *et al.* 2013). However, such problems characterized by heterogeneous agents, interrelated utilities and complex intertask dependencies are typically NP hard. Therefore, the current challenges faced are to establish an effective model to describe the cooperative task planning of heterogeneous MRSs and develop a practical method to provide acceptable solutions for practical large-size problems.

In recent years, significant progress has been made in describing the cooperative task planning of heterogeneous MRSs. Specifically, two commonly used taxonomies describing the characteristics between robots, tasks, time and constraints have been established. The first taxonomy was proposed by Gerkey and Mataric (2004) to describe multi-robot task allocation (MRTA) problems through three main dimensions: (1) single-task robots (ST) versus multi-task robots (MT); (2) single-robot tasks (SR) versus multi-robot tasks (MR); and (3) instantaneous assignment (IA) versus time-extended assignment (TA). Therefore, the cooperative task planning of heterogeneous MRSs belongs to the ST-MR-IA category of MRTA, which often appears in real-world applications where some tasks require the combined effort of multiple robots. Because of the increasing importance of interrelated utilities and constraints in realistic MRTA problems, a new comprehensive taxonomy was presented by Korsah, Stentz, and Dias (2013), which explicitly takes the issues of interrelated utilities and constraints into consideration. Under the new classification framework, the problem can

be viewed as the ST-MR-IA problem with complex dependencies (CD [ST-MR-IA]). However, few studies have described these problems in detail and systematically.

The most commonly used approaches for solving MRTA problems can be categorized into behaviour-based approaches, negotiation-based approaches and optimization-based approaches. Parker (1998) developed a fully distributed behaviour-based software architecture which gives all robots the capability to determine their own actions based on their current situations. The behaviour-based approaches do not support explicit communication between agents to allocate the tasks, whereas the negotiation-based approaches can determine task assignments through individual communication between agents. Since different assignments with the same coalition can have different utilities, most algorithms from the agent-based coalition formation problem cannot be directly applied to the ST-MR-IA problem (Zhang and Parker 2013).

To theoretically obtain the optimal scheme of task allocation, researchers have focused more extensively on optimization-based approaches (Khamis, Hussein, and Elmogy 2015). Jones, Dias, and Stentz (2011) utilized the genetic algorithm (GA) to solve the time-extended task allocation of multiple robots in a simulated disaster scenario. Wang, Gu, and Li (2012) designed the ant colony optimization (ACO) algorithm for optimizing the task allocation problem of MRSs. However, optimization-based approaches for optimizing cooperative task planning of heterogeneous MRSs have not been systematically researched. To fill the above research gaps, this article establishes a novel mapping mechanism which describes the cooperative task planning of heterogeneous MRSs in an order picking warehouse as OSS-SSTT, and then develops an effective HABC to increase the size of solvable problems and improve the performance of the solution.

3. Problem formulation

This study considers a class of problems in which complex tasks are allocated to a group of heterogeneous multi-robots with two specialized capabilities. Specifically, in an order picking warehouse, the multi-station pick-up and delivery tasks require one robot to travel between different stations in multiple workshops, and each workshop is equipped with a mobile manipulator (*e.g.* worker) or a fixed manipulator (*e.g.* robot) to help the arriving robot to load items. In the rest of this section, the cooperative task planning of heterogeneous MRSs in an order picking warehouse is described and transformed into the well-known OSS-SSTT, and then an efficient MILP model is developed for OSS-SSTT.

3.1. Problem description

The cooperative task planning problem of heterogeneous MRSs in an order picking warehouse refers to the arrangement for a group of cooperative robots to fetch specific items in different locations under the help of manipulators. A simple case of heterogeneous MRSs in an order picking warehouse is shown in Figure 1. In this case, the entire workspace is partitioned into four workshops, and each workshop is equipped with one worker. In each workshop, four stations represent the acquisition locations of different products. The products at different stations in a workshop are different, but they are consistent among different workshops. In the workspace, the robot's mission is to move to one specified station in each workshop in a predetermined order without disturbing the path of other robots, and load products with the help of workers. The worker has to move from one station to another in a group of stations belonging to different robot acquisition locations to help the robot to load products in one workshop. The problem is to find an optimal task planning of heterogeneous MRSs with the shortest completion time.

Table 1 shows the detailed mapping relationship between this problem and OSS-SSTT. A robot acquiring items in each workshop is regarded as a job, in which a robot acquiring items with the help of a worker at a specified location in one workshop is viewed as an operation of the job. The worker in one workshop is responsible for helping the arriving robots to load items; and this is treated as

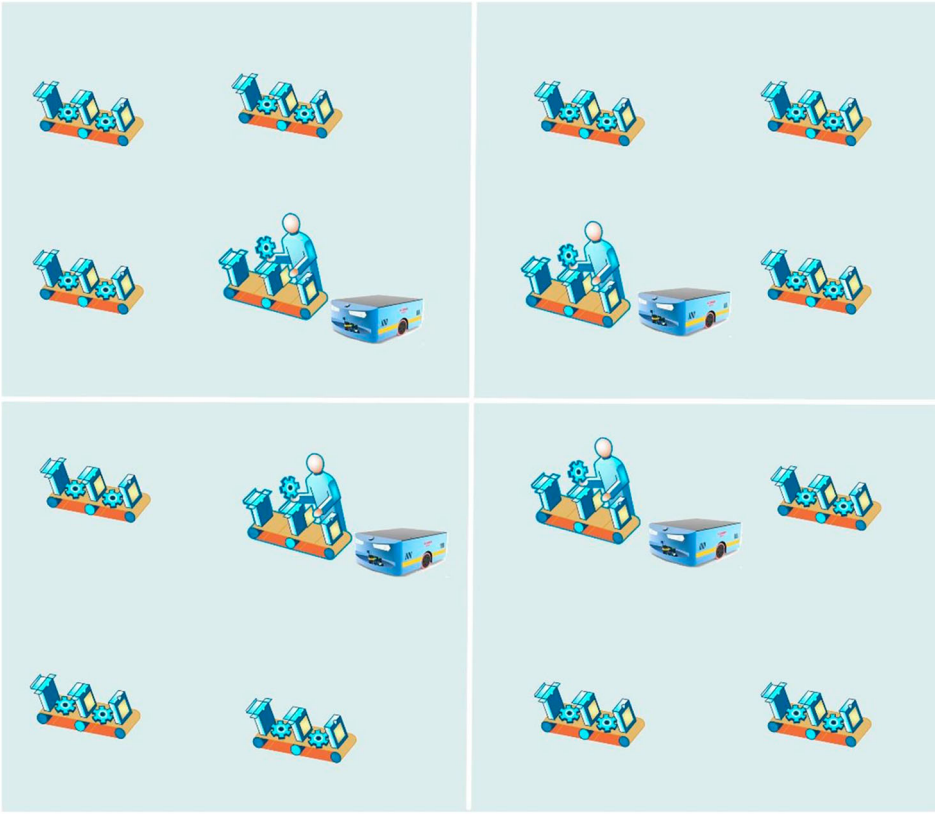


Figure 1. Layout of an order picking warehouse.

Table 1. Mapping between cooperative task planning of heterogeneous multi-robot systems (MRSs) and open shop scheduling with sequence-dependent set-up and transportation times (OSS-SSTT).

Cooperative task planning of heterogeneous MRSs	OSS-SSTT
Robot with allocated tasks	Job
Robot with one subtask	Operation
Worker	Machine
Execution time of the subtask	Processing time
Travelling time of the robot	Transportation time
Switching time of the worker	Set-up time
Task plans for robots	Sequences of jobs
Task plans for workers	Sequences on machines

a machine. The time required for a robot to load items with the help of a worker is the processing time. The time required for a robot to travel from one specified location in a workshop to another in the next workshop is the transportation time, and the time required for a worker to move from one specified location to another within a workshop is the set-up time. Through these mapping rules, the cooperative task planning problem of heterogeneous MRSs is completely equivalent to OSS-SSTT.

Now, consider a set containing n robots $J = \{J_1, J_2, \dots, J_n\}$, and a set containing m workers $M = \{M_1, M_2, \dots, M_m\}$. Robots travel across different stations, where the operation $O_{j,i}$ is performed by the j th robot under the help of the i th worker in the i th workshop, with a fixed processing time $P_{j,i}$. One robot performs at most m operations in total and at most an operation in each workshop. Each worker performs at most n operations in one workshop. At any time, one worker and one

Table 2. Parameters and indices used in the model.

Notation	Description
n	Number of jobs
m	Number of machines
$O_{j,i}$	Operation of job j on machine i
$P_{j,i}$	Processing time of $O_{j,i}$
Q	A large positive number
j, k	Index for job $\{1, 2, \dots, n\}$
i, l	Index for machines $\{1, 2, \dots, m\}$
$IST_{j,i}$	Initial set-up time for $O_{j,i}$
$JST_{j,i,l}$	Delivery time between $O_{j,i}$ and $O_{j,l}$ for job j
$MST_{j,i,k}$	Set-up time between $O_{k,i}$ and $O_{j,i}$ on machine i

robot can perform at most one operation. For simplicity without loss of generality, it is assumed that both the robots and the workers move at unit speed. Then, based on the coordinate location array, the distance between two loading locations $[(x_{j,i}, y_{j,i}) \text{ and } (x_{j,l}, y_{j,l})]$ of a robot J_j is regarded as $JST_{j,i,l}$, which can be calculated by Equation (1):

$$JST_{j,i,l} = |x_{j,i} - x_{j,l}| + |y_{j,i} - y_{j,l}| \quad (1)$$

The distance between two loading locations $[(x_{j,i}, y_{j,i}) \text{ and } (x_{k,i}, y_{k,i})]$ of a worker M_i in a workshop is considered as $MST_{j,i,k}$, which can be calculated by Equation (2):

$$MST_{j,i,k} = |x_{j,i} - x_{k,i}| + |y_{j,i} - y_{k,i}| \quad (2)$$

The initial set-up times of robots (IJST) and workers (IMST) are calculated by Equations (3) and (4), respectively, based on the initial position coordinates of robots (x_{rpj}, y_{rpj}) and workers (x_{wpi}, y_{wpi}) .

$$IJST_{j,i} = |x_{j,i} - x_{rpj}| + |y_{j,i} - y_{rpj}| \quad (3)$$

$$IMST_{j,i} = |x_{j,i} - x_{wpi}| + |y_{j,i} - y_{wpi}| \quad (4)$$

The initial set-up times (IST) can be calculated by Equation (5). Then, the cooperative task planning of heterogeneous MRSSs is completely consistent with OSS-SSTT:

$$IST_{j,i} = \max(IJST_{j,i}, IMST_{j,i}) \quad (5)$$

3.2. MILP formulation

To obtain the optimal scheme for cooperative task planning of heterogeneous MRSSs, a MILP model is developed to formulate the OSS-SSTT. Before presenting the model, the sets and parameters used are listed in Table 2, and the following assumptions are made:

- (1) All jobs are at the beginning of the planning horizon.
- (2) At any time, a job can be processed by at most one machine, and one machine can process only one job.
- (3) Processing times are known in advance.
- (4) Pre-emption of operations is not allowed.
- (5) Stochastic uncertain events such as machine failure are not considered.

To reduce the required number of binary variables, the definitions of variables x and y used in this model are consistent with Naderi *et al.* (2011). The detailed definitions of decision variables are as follows:

$S_{j,i}$: starting time of operation $O_{j,i}$.
 $C_{\max}$: completion time of the last job.

$$y_{j,i,k} = \begin{cases} 1 & \text{if } O_{j,i} \text{ is processed after } O_{k,i} \text{ (not necessarily immediately)} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{j,i,l} = \begin{cases} 1 & \text{if } O_{j,i} \text{ is processed after } O_{j,l} \text{ (not necessarily immediately)} \\ 0 & \text{otherwise} \end{cases}$$

The MILP model for OSS-SSTT is as follows:

$$\min C_{\max} \quad (6)$$

subject to:

$$S_{j,i} \geq IST_{j,i} \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m\} \quad (7)$$

$$C_{\max} \geq S_{j,i} + P_{j,i} \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m\} \quad (8)$$

$$S_{j,i} \geq S_{j,l} + P_{j,l} + JST_{j,i,l} - Q \cdot (1 - x_{j,i,l}) \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m-1\}, l > i \quad (9)$$

$$S_{j,i} \geq S_{j,i} + P_{j,i} + JST_{j,i,l} - Q \cdot x_{j,i,l} \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m-1\}, l > i \quad (10)$$

$$S_{j,i} \geq S_{k,i} + P_{k,i} + MST_{j,i,k} - Q \cdot (1 - y_{j,i,k}) \quad \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, n-1\}, k > j \quad (11)$$

$$S_{k,i} \geq S_{j,i} + P_{j,i} + MST_{j,i,k} - Q \cdot y_{j,i,k} \quad \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, n-1\}, k > j \quad (12)$$

$$S_{j,i} \geq 0 \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m\} \quad (13)$$

$$x_{j,i,l} \in \{0, 1\} \quad \forall j \in \{1, 2, \dots, n\}, i \in \{1, 2, \dots, m-1\}, l > i \quad (14)$$

$$y_{j,i,k} \in \{0, 1\} \quad \forall i \in \{1, 2, \dots, m\}, j \in \{1, 2, \dots, n-1\}, k > j \quad (15)$$

The objective function (6) minimizes the completion time of the last job to leave the system. Constraint set (7) indicates that an operation cannot be processed earlier than its initial set-up time. Constraint set (8) states that the makespan is no earlier than the completion time of any operation. Constraint sets (9) and (10) are pairs of constraints implying the precedence of two operations of a job. Similarly, constraint sets (11) and (12) are pairs of constraints implying the precedence of two different jobs on a machine. Constraint sets (13), (14) and (15) define the decision variables.

4. Solution approach

Although the cooperative task planning optimization problem of heterogeneous MRSs has not been systematically researched, the open shop scheduling problem has received considerable attention from researchers around the world since it was first raised by Gonzalez and Sahni (1976). Many metaheuristic algorithms, such as the GA (Abreu *et al.* 2020), particle swarm optimization (PSO) (Sha and Hsu 2008) and cuckoo search (CS) (Huang *et al.* 2019), have been used to solve the open shop scheduling problem. However, the effectiveness of these metaheuristic algorithms in solving OSS-SSTT is still under investigation.

As a new algorithm that mimics the foraging behaviour of bees, the artificial bee colony (ABC) algorithm is regarded as a pioneer of swarm intelligence algorithms (Shukla and Tiwari 2017). In the ABC algorithm, a food source represents a solution designed for the problem, and there are three kinds of bees in the bee colony: employed bees, onlooker bees and scout bees. Employed bees collect

food in specific food sources and change into scout bees when the food source is exhausted. Scout bees randomly search for new food sources and change into employed bees linked to the new food source. Onlooker bees gather food at certain food sources based on the food density of all food sources. In short, employed bees and onlooker bees perform local searches while scout bees perform global searches, thus achieving a good balance between exploration and exploitation. The ABC algorithm has been widely applied in various optimization problems and achieved good results (Karaboga *et al.* 2014; Li *et al.* 2018; Luo 2020).

During each iteration of the ABC algorithm, local search and global search are performed simultaneously, which gives the algorithm a certain ability to quickly converge and escape from the local optimal solution. However, with the increase in the problem size, the complexity of the problem increases rapidly, which makes the inherent processing capacity of the algorithm insufficient. Therefore, this study improves the basic ABC algorithm by introducing several concepts from the tabu search (TS) algorithm and artificial immune (AI) algorithm, and develops an HABC for OSS-SSTT.

The general idea of the proposed HABC algorithm is as follows. First, initial solutions called food sources are generated randomly in the solution space. The fitness of every solution is calculated and the best solution is chosen. Next, employed bees perform the local search. Then, onlooker bees choose food sources according to the fitness of food sources and perform another round of the local search. Finally, scout bees are generated and implement the global search when food sources are exhausted. The process runs repeatedly until the termination condition is reached. The detailed HABC algorithm is outlined in Figure 2. Some important modules of the algorithm are described in detail as follows.

4.1. Encoding rule

After comprehensive consideration of the length and range of the code and the difficulty of decoding, the encoding scheme described in Hosseinabadi *et al.* (2019) is adopted. In the encoding process, a permutation sequence from 1 to $n*m$ is utilized to determine the processing order of all operations for OSS-SSTT with n jobs and m machines.

4.2. Employed bees

In each generation of the HABC algorithm, employed bees search for food sources associated with them. Then, each employed bee carries out one local search, which slightly changes the sequence of food sources and generates a new food source. If the new food source is better, this food source will be updated. By this means, the quality of the food source can be improved. To enhance the search capability of the local search, a combinatorial random local search method is designed in this study. As shown in Figure 3, four distinct local search methods are grouped, and one of them will be randomly chosen to perform one local search when needed. The first method is flipping a sub-sequence of the solution, while the length and position of the sub-sequence are randomly generated to enlarge the scope of the local search. The idea of the second method comes from the bit shift in the computer system, using a randomly generated number that does not exceed the length of the solution sequence as the step size of the sequence shifting to the right. In the third method, a new solution is generated by swapping the chosen two elements. In the final method, only a part of the solution sequence is shifted and the shift step is one.

However, during the search process, there is a situation in which a new solution is generated repeatedly. This prevents the algorithm from making an effective local search to find a better solution, wasting a lot of time. Therefore, when the local search finds a better solution at the neighbour of the first solution, a tabu table is introduced to allow the local search to jump out of the first solution and regenerate a new neighbour solution to accelerate the search process, as shown in Figure 4.

Algorithm 1: HABC algorithm

```

1: Randomly generate initial food sources  $x_i$  ( $i = 1, 2, \dots, P$ ), and assign them to employed bees
2: Calculate the quality of each food sources  $f(x_i)$ ,  $i = 1, 2, \dots, P$ 
3: Initialize the times  $l_i = 0$  ( $i = 1, 2, \dots, P$ ) without improvement
4: Initialize tabu table  $T_i = 0$  ( $i = 1, 2, \dots, P$ ) for each food source
5: while stopping criteria is not satisfied do
6:   Calculate the best food source  $x_{best}$ 
7:   //Employed bee search process
8:   for  $i \leq P$  do
9:      $x_{new} \leftarrow \text{Localsearch}(x_i)$ 
10:    if  $x_{new}$  is not in  $T_i$  then
11:      if  $f(x_{new})$  is better than  $f(x_i)$  then
12:         $x_i \leftarrow x_{new}$ ;  $l_i \leftarrow 0$ 
13:        Add  $x_i$  into tabu table  $T_i$ 
14:      else  $l_i \leftarrow l_i + 1$ ;
15:      end if
16:    end if
17:  end for
18:  //Onlooker bee search process
19:  Calculate the probability of each food source  $x_i$  based on  $f(x_i)$ 
20:  for  $i \leq P$  do
21:    Select a food source  $x_i$  using Roulette wheel selection method
22:     $x_{new} \leftarrow \text{Localsearch}(x_i)$ 
23:    if  $x_{new}$  is not in  $T_i$  then
24:       $x_i \leftarrow x_{new}$ ;  $l_i \leftarrow 0$ 
25:      Add  $x_i$  into tabu table  $T_i$ 
26:    end if
27:  end for
28:  //Scout bee search process
29:  for  $i \leq P$  do
30:    if  $l_i = \text{limit}$  then
31:       $x_i$  is replaced by a randomly generated new food source
32:       $T_i \leftarrow 0$ ;  $l_i \leftarrow 0$ 
33:    end if
34:  end for
35:  for every two food sources  $x_i$  and  $x_j$  do
36:    if similarity of  $x_i$  and  $x_j$  reaches threshold then
37:       $x_i$  is replaced by a randomly generated new food source
38:       $T_i \leftarrow 0$ ;  $l_i \leftarrow 0$ 
39:    end if
40:  end for
41:  Introduce  $d$  new food sources
42:  Abandon  $d$  worst food sources to compress population
43: end while

```

Figure 2. Pseudo-codes of the hybrid artificial bee colony (HABC) algorithm.

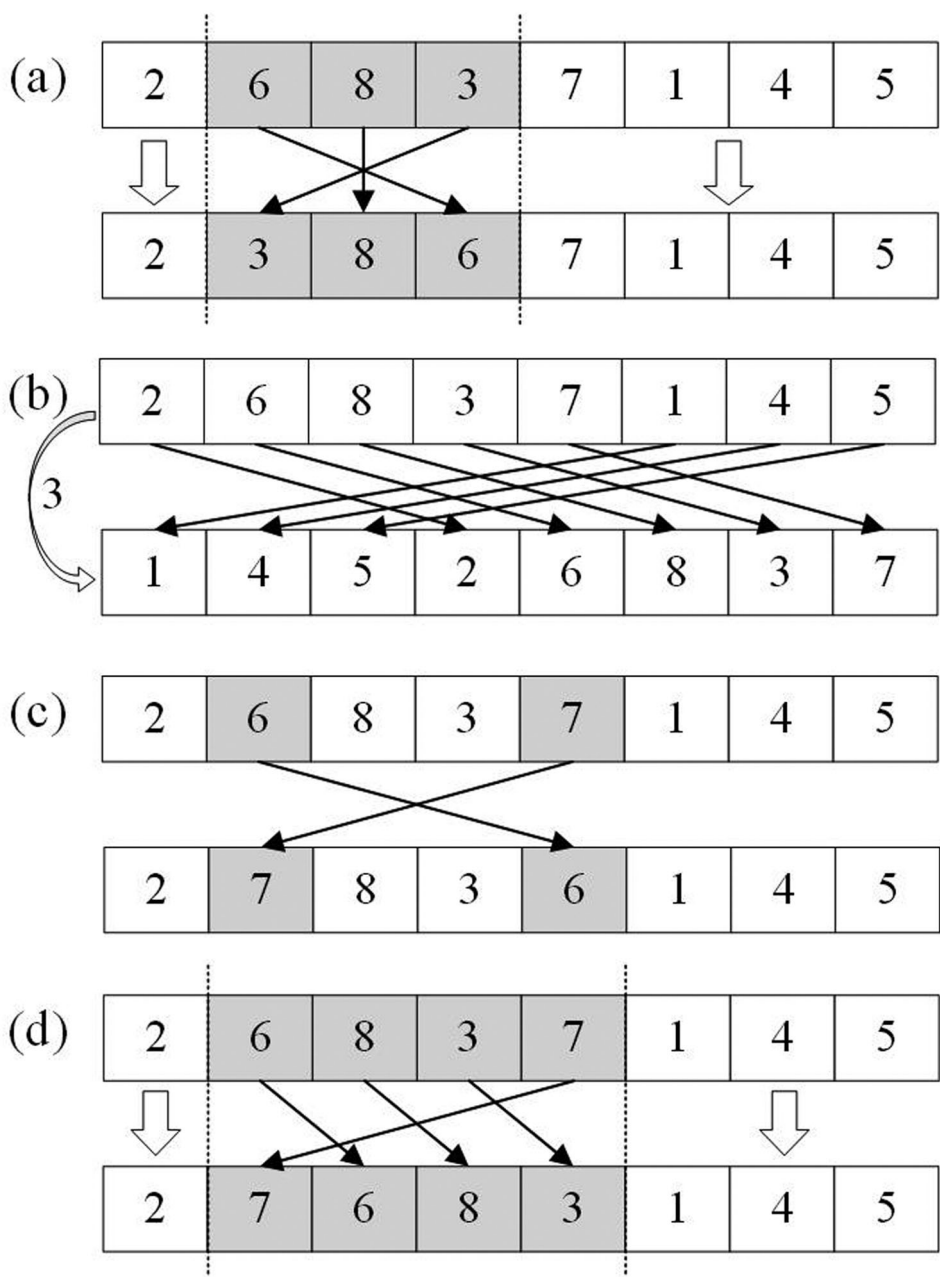


Figure 3. Examples of local search method.

4.3. Onlooker bees

After the search process of employed bees, onlooker bees search for better results to further improve the quality of food sources. To maximize the effect of the onlooker search, the quality of all food sources is first evaluated, and then the food source with good quality will be selected for the search with a higher probability using a roulette wheel selection process. The subsequent local search method and updating process are the same as those used by employed bees.

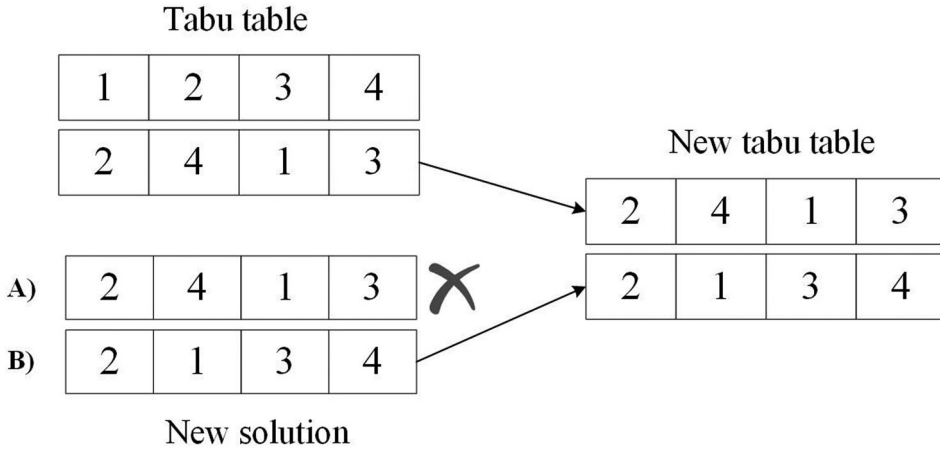


Figure 4. Example of tabu table update procedure.

4.4. Scout bees

In the HABC algorithm, a global search is performed by each scout bee in every generation. Scout bees are generated from employed bees. When the employed bee performs one local search, a counter is used to count the number of invalid local search results. If an employed bee has not found a better solution after performing a fixed number of local searches, the food source will be regarded as an exhausted food source and the employed bee will turn into a scout bee. Then, the scout bee performs a global search by generating a random solution as a new food source in the solution space and turns back into an employed bee to perform a local search again.

The gradual improvement in the solution quality is achieved through efficient local and global searches during a large number of iterations. However, since these generated solutions often become similar after several generations and approach the same local optima, the quality of the solution will be difficult to improve fundamentally, even if a lot of time is spent in continuing the search. To avoid this situation, the similarity avoiding process of the AI algorithm is utilized (Lanaridis and Stafylopatis 2014). In the HABC algorithm, the similarity of all individuals in each generation will be calculated using $s = n/m$, where n is the number of the same elements within two solutions and m is the length of the solution.

Then, if the similarity is higher than the previously given threshold, only one individual will be retained and the other will be replaced by a randomly generated solution. As shown in Figure 5, the first two individuals are not very similar, so both of them can be preserved, while the last two are similar and the last one is replaced with a new solution.

Beyond that, since the randomly generated solutions may be in a poor position in the solution space, the previously designed local and global search cannot achieve an acceptable solution even if these searches consume more computational resources and computing time. To overcome this deficiency, the population refresh process is introduced as well. Specifically, in each generation of the HABC algorithm, a certain number of bad individuals will be replaced with the randomly generated solutions to enhance the search ability of the algorithm.

5. Computational experiments

To investigate the effectiveness and performance of the proposed HABC algorithm for cooperative task planning of heterogeneous multi-robot systems in an order picking warehouse, three groups of experiments are considered. The first group of experiments is conducted to determine the appropriate parameter setting for the HABC algorithm. In the second group of experiments, problems of different

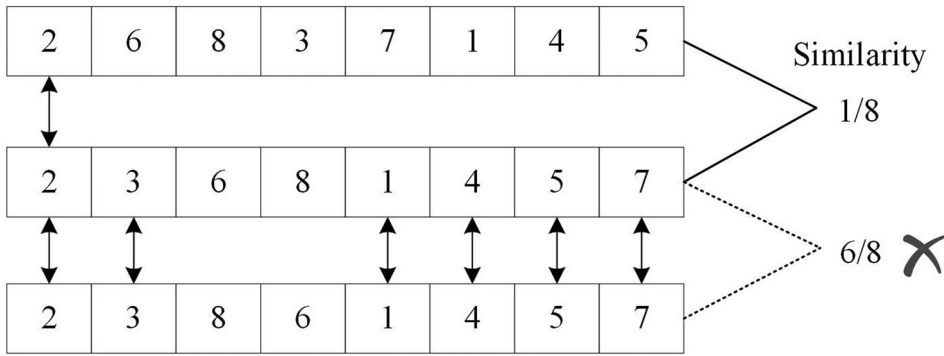


Figure 5. Similarity avoiding process.

size with a fixed number of workshops are tested. The third group of experiments is performed with a variable number of workshops. The experimental results of the presented HABC are compared with the state-of-the-art algorithms reported in the literature. The MILP model is solved using the CPLEX 12.9 solver in the MATLAB R2019a environment, and all the experiments are implemented on a personal computer with an i7-8700 central processing unit at 3.2 GHz and with 16 GB RAM.

5.1. Parameter setting of the HABC

Since the performance of the metaheuristic algorithm depends greatly on the value of the algorithm parameters, parameter setting of the specific metaheuristic plays a crucial role in approaching the limit of its potential. When the algorithm has multiple undetermined parameters with a considerable number of values and each experiment takes a long time, the design of experiment technique is essential for numerical experiments (Liu, Zhang, and Lu 2019). In the proposed HABC, there are five main parameters to be determined: population scale (*scale*), local search limit (*limit*), length of tabu table (*tabusize*), similarity threshold (*simthres*) and population refresh ratio (*refratio*). In this research, each factor has four levels: *scale* (25, 50, 75, 100), *limit* (5, 10, 15, 20), *tabusize* (10, 20, 30, 40), *simthres* (0.6, 0.7, 0.8, 0.9) and *refratio* (0.05, 0.1, 0.15, 0.2). The selection of these levels is based on several trials under different settings and other articles. To reduce the number of runs but reach sound conclusions, an orthogonal experiment with size L16 (4^5) is used to calibrate these parameters. These 16 groups of parameters are denoted by Com1, Com2, . . . , Com16, respectively, and experimental results are shown in Table 3.

As shown in Figure 6, an order picking warehouse scenario consisting of nine workshops is created, where each workshop has nine stations and is equipped with a worker. Initial positions of stations and workers are randomly generated and represented by solid points and hollow circles, respectively. Moreover, robots are initially set at the left corner of each workshop. An operation can start only when both the related robot and worker reach the station.

In the experiment, each group of parameters is tested 10 times and the stopping criterion is the elapsed time of 1800 s. The makespan objective is used to evaluate the result. Table 3 shows that under the seventh combination of (50, 15, 40, 0.7, 0.05), the Avg is the smallest, and thus Com7 is the best. However, when these parameters are investigated independently in Figure 7, it is obvious that the best *scale* is 50, the best *limit* is 15, the best *simthres* is 0.8 and the best *refratio* is 0.05. When it comes to *tabusize*, since the curve rises first and then falls, a better *tabusize* is either 10 or 40. The above results are not completely consistent with that of the seventh combination. Considering the incompleteness of the orthogonal experiment, two more tests of the combinations of (50, 15, 10, 0.8, 0.05) and (50, 15, 40, 0.8, 0.05) are performed, and their Avg values are 2543.6 and 2547.8, respectively. Taking these 18 experiments into account, it can be concluded that (50, 15, 40, 0.7, 0.05) is the optimal parameter

Table 3. Parameter settings of the orthogonal experiment.

Com	Level					Avg
	scale	limit	tabusize	simthres	refratio	
1	25	5	10	0.6	0.05	2636.8
2	25	10	20	0.7	0.1	2599.7
3	25	15	30	0.8	0.15	2555.9
4	25	20	40	0.9	0.2	2566.0
5	50	5	20	0.9	0.15	2657.5
6	50	10	10	0.8	0.2	2585.5
7	50	15	40	0.7	0.05	2524.9
8	50	20	30	0.6	0.1	2532.2
9	75	5	30	0.7	0.2	2655.1
10	75	10	40	0.6	0.15	2589.2
11	75	15	10	0.9	0.1	2549.8
12	75	20	20	0.8	0.05	2570.9
13	100	5	40	0.8	0.1	2646.6
14	100	10	30	0.9	0.05	2593.3
15	100	15	20	0.6	0.2	2603.3
16	100	20	10	0.7	0.15	2580.9

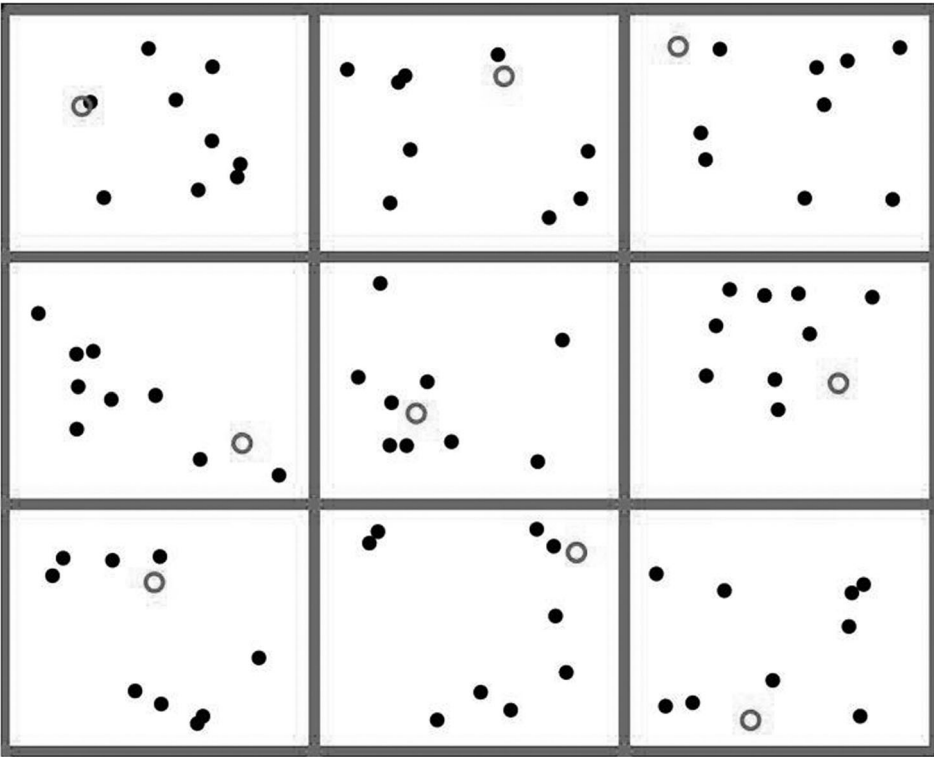


Figure 6. Warehouse scenario with nine workshops and nine robots.

combination for population scale, local search limit, length of tabu table, similarity threshold and population refresh ratio.

5.2. Experimental results with a fixed number of workshops

As mentioned above, few studies have dealt with cooperative task planning of heterogeneous multi-robot systems in an order picking warehouse. Owing to the lack of benchmark instances and

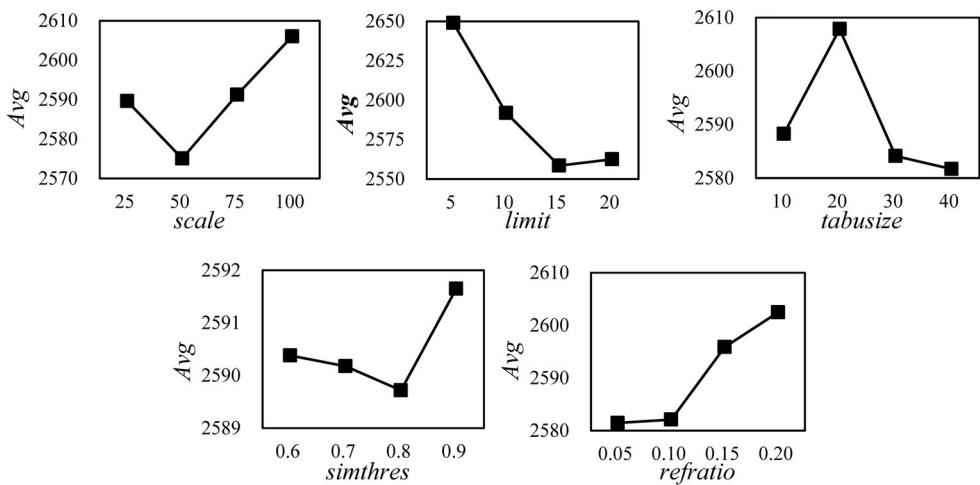


Figure 7. Estimated marginal means of Avg on five parameters.

Table 4. Results of experiments with a fixed number of workshops.

No.	Workshop	Robot	$t_{\max}$ (min)	MILP	GA	PSO	CS	ABC	ABC-AI	ABC-TS	HABC
1	2*2	4	8	917.46	927.02	977.88	946.32	926.15	925.96	927.05	919.87
2	2*2	5	10	1022.4	1082.2	1109.1	1093.2	1051.7	1050.4	1058.6	1051.3
3	2*2	6	12	1166.8	1221.2	1254.7	1250.7	1217.2	1214.4	1208.3	1206.2
4	2*2	7	14	1344.5	1398.9	1410.3	1424.5	1388.0	1387.3	1380.8	1379.7
5	2*2	8	16	1467.4	1542.3	1567.7	1586.1	1541.4	1541.2	1536.9	1531.5
6	2*2	9	18	1599.4	1675.1	1706.4	1727.7	1691.7	1690.5	1680.2	1669.4
7	2*2	10	20	1896.4	1926.3	1968.6	1998.1	1953.8	1947.2	1948.1	1931.3
8	2*2	15	25	2742.1	2830.5	2841.1	2899.8	2812.9	2803.5	2810.4	2799.8
9	2*2	20	30	3624.4	3627.6	3662.2	3775.0	3629.0	3622.8	3619.4	3597.5
10	2*2	25	35	4478.6	4530.7	4581.8	4693.2	4507.9	4492.7	4487.5	4452.2
11	2*2	30	40	5379.5	5412.6	5479.3	5604.0	5394.7	5358.7	5370.2	5320.1
12	2*2	35	45	6180.4	6173.2	6360.6	6458.2	6209.4	6170.1	6169.8	6132.5
13	2*2	40	50	7151.0	7189.7	7374.0	7477.8	7156.0	7127.3	7117.5	7062.3
14	2*2	45	55	8197.1	8011.7	8262.4	8296.7	7981.0	7953.8	7955.2	7891.4
15	2*2	50	60	9051.3	8812.9	9139.6	9177.2	8829.8	8773.5	8783.6	8713.7

Note: MILP = mixed-integer linear programming; GA = genetic algorithm; PSO = particle swarm optimization; CS = cuckoo search; ABC = artificial bee colony; ABC-AI = artificial bee colony with artificial immune algorithm; ABC-TS = artificial bee colony with tabu search; HABC = hybrid artificial bee colony. Bold number represents the smallest average makespan of all algorithms in this instance.

comparison algorithms, the artificial bee colony algorithm with artificial immune (ABC-AI) and artificial bee colony algorithm with tabu search (ABC-TS) are compared with the ABC algorithm to illustrate the superiority of introducing the artificial immune and tabu search algorithms, respectively, into the ABC algorithm. Then, the HABC algorithm is compared with the above three algorithms to validate the superiority and necessity of considering AI and TS simultaneously in the ABC algorithm. It is worth noting that ABC-AI, ABC-TS and HABC have the same structure as ABC and are formed by adding, respectively, AI, TS, and both of them. Finally, to investigate the superiority of the proposed algorithm compared with other algorithms, it is compared with the exact method and other metaheuristic methods, namely GA, PSO and CS.

To study the effectiveness of these algorithms for problems of different size with a fixed number of workshops, experiments in an order picking warehouse scenario with four workshops and four to 50 robots are carried out. As shown in Table 4, $t_{\max}$ is the maximum CPU time of the algorithm, which is determined in proportion to the problem size. In addition, the result is an average makespan of 10 repeated runs.

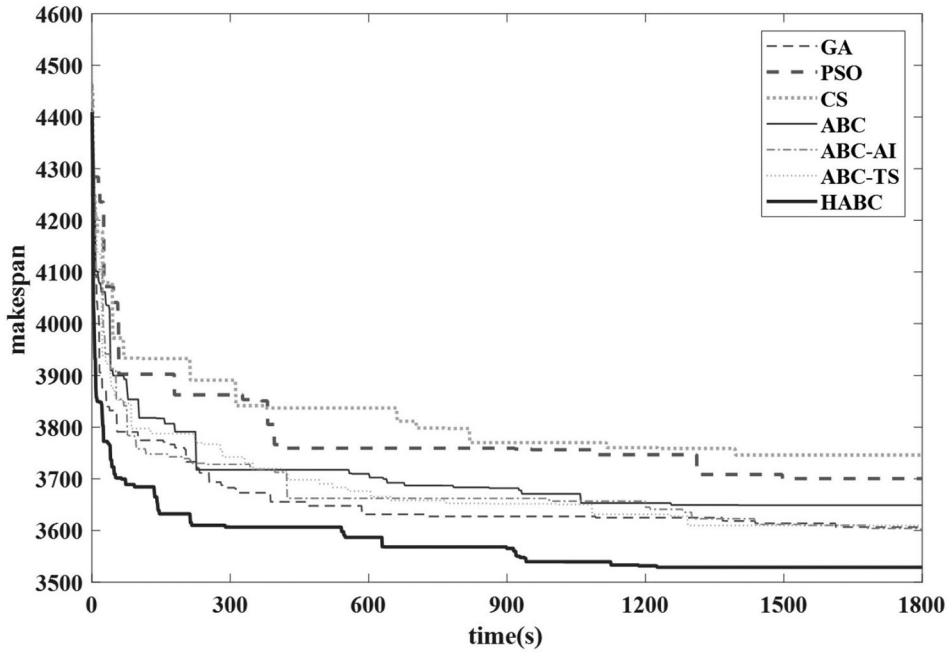


Figure 8. Evolution process of various metaheuristic algorithms. GA = genetic algorithm; PSO = particle swarm optimization; CS = cuckoo search; ABC = artificial bee colony; ABC-AI = artificial bee colony with artificial immune algorithm; ABC-TS = artificial bee colony with tabu search; HABC = hybrid artificial bee colony.

To figure out the convergence speeds of the proposed algorithm, Figure 8 shows the convergence curve of all algorithms for the instance with four workshops and 20 robots. It is obvious from Figure 8 that the proposed HABC has the fastest convergence speed and best solution. Moreover, the fitness values continuously decrease during the whole process, which proves the necessity of the long CPU time.

To more intuitively demonstrate the effectiveness of these three metaheuristic algorithms (ABC-AI, ABC-TS and HABC) compared with ABC, the increased percentage of the results obtained by these algorithms is calculated by Equation (16), where X refers to one of these three algorithms and $T(X)$ refers to the average makespan obtained by X . Given that a lower makespan means a better solution, the lower $INC1$ means better algorithm performance.

$$INC1 = \frac{T(X) - T(ABC)}{T(ABC)} \quad (16)$$

As shown in Figure 9, it is evident that both ABC-AI and ABC-TS have achieved about 0.5% improvement over ABC, which proves the effectiveness of considering AI and TS, respectively, based on the basic ABC algorithm. Besides, HABC makes significant further improvements compared with ABC-AI and ABC-TS, and achieves over 1.3% improvement over ABC. Moreover, the improvement of HABC compared to ABC becomes more significant as the problem size increases. This fully demonstrates the necessity and effectiveness of introducing AI and TS simultaneously into the ABC algorithm.

Similarly, the increased percentage of the results obtained by metaheuristics compared to the MILP solution is calculated using Equation (17) to better demonstrate the effectiveness of HABC and other metaheuristics compared to MILP.

$$INC2 = \frac{T(X) - T(MILP)}{T(MILP)} \quad (17)$$

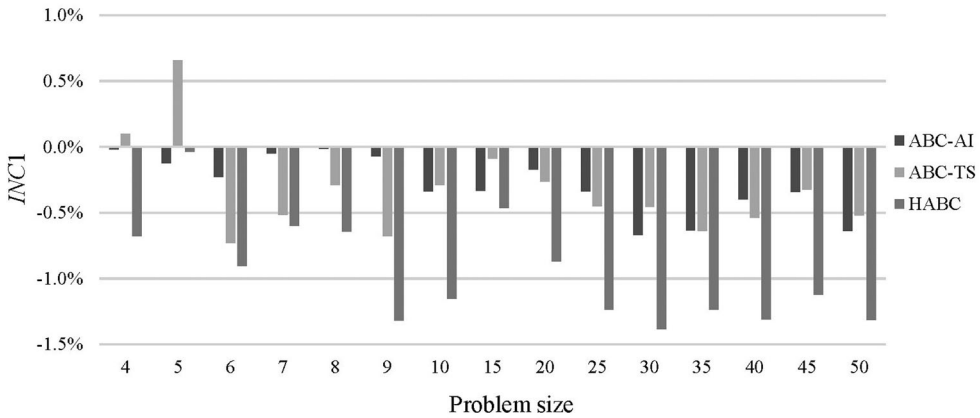


Figure 9. Results of hybrid algorithms compared with the artificial bee colony (ABC) algorithm for experiments with a fixed number of workshops. *INC1* = increased percentage; ABC-AI = artificial bee colony with artificial immune algorithm; ABC-TS = artificial bee colony with tabu search; HABC = hybrid artificial bee colony.

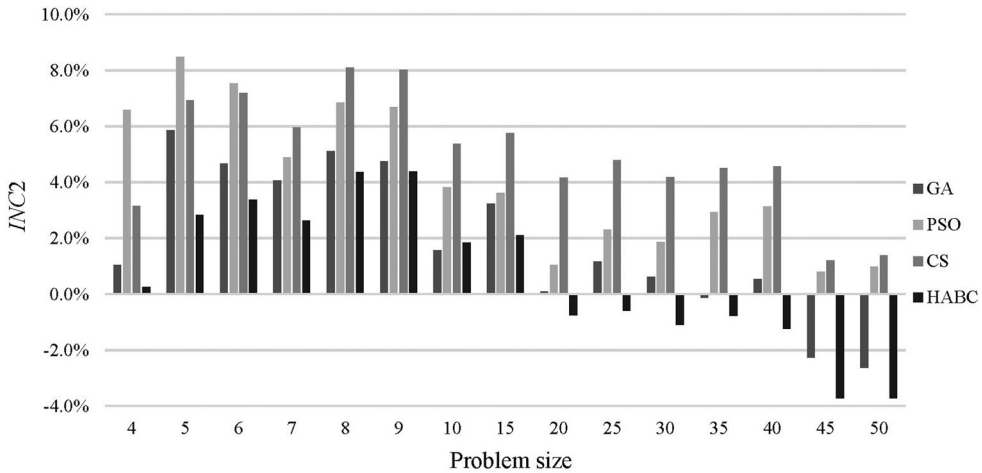


Figure 10. Results of metaheuristic algorithms compared with mixed-integer linear programming (MILP) for experiments with a fixed number of workshops. *INC2* = increased percentage; GA = genetic algorithm; PSO = particle swarm optimization; CS = cuckoo search; HABC = hybrid artificial bee colony.

As can be seen from Figure 10, the results obtained by all metaheuristic algorithms in the same amount of time present a trend of improvement compared to MILP as the problem size increases. For the exact method, since the maximum CPU time is limited, the algorithm output will be the current best solution of the branch-and-bound method and may not be the global optima. In particular, when the number of robots exceeds 20, the results obtained by HABC are superior to those of MILP in the same computing time. In addition, HABC can achieve better results than other metaheuristic algorithms on almost any problem size, which further demonstrates the effectiveness of the proposed algorithm.

5.3. Experimental results with a variable number of workshops

Experiments in an order picking warehouse scenario with a variable number of workshops are carried out to further investigate the effectiveness of the proposed algorithm compared to other metaheuristic

Table 5. Results of experiments with a variable number of workshops.

No.	Workshop	Robot	$t_{\max}$ (min)	GA	PSO	CS	ABC	ABC-AI	ABC-TS	HABC
1	2*3	6	20	1527.5	1547.6	1600.2	1522.8	1528.3	1521.3	1493.1
2	2*4	8	30	2246.3	2344.7	2473.2	2307.1	2328.7	2299.0	2224.7
3	3*3	9	45	2548.6	2746.0	2962.9	2635.9	2613.5	2618.0	2528.8
4	3*4	12	50	3897.5	4451.9	4594.4	3926.4	3856.4	3904.1	3727.2
5	3*5	15	55	5436.4	6500.3	6726.1	5549.4	5415.6	5438.4	5241.3
6	4*4	16	60	5852.2	7052.4	7407.4	5928.3	5841.4	5805.7	5600.3

Note: GA = genetic algorithm; PSO = particle swarm optimization; CS = cuckoo search; ABC = artificial bee colony; ABC-AI = artificial bee colony with artificial immune algorithm; ABC-TS = artificial bee colony with tabu search; HABC = hybrid artificial bee colony. Bold number represents the smallest average makespan of all algorithms in this instance.

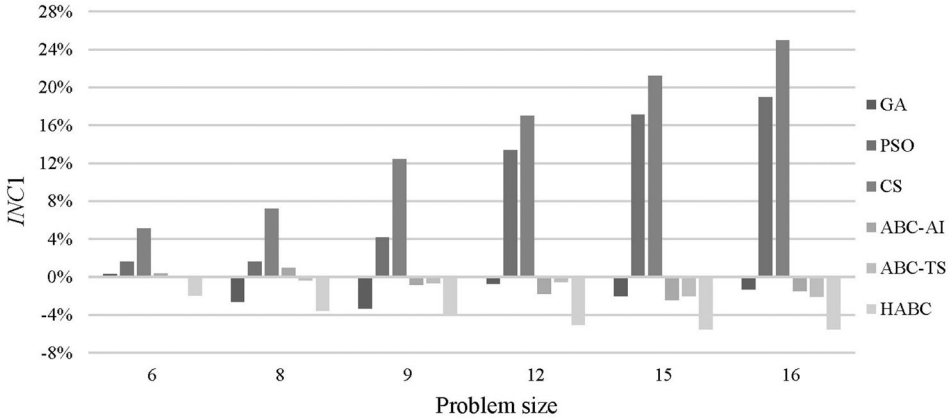


Figure 11. Results of all other algorithms compared with the artificial bee colony (ABC) for experiments with a variable number of workshops. $INC1$ = increased percentage; GA = genetic algorithm; PSO = particle swarm optimization; CS = cuckoo search; ABC = artificial bee colony; ABC-AI = artificial bee colony with artificial immune algorithm; ABC-TS = artificial bee colony with tabu search; HABC = hybrid artificial bee colony.

algorithms. The results are shown in Table 5. To better illustrate the effectiveness of all algorithms for cooperative task planning of heterogeneous multi-robot systems, the increased percentage of the results obtained by all other algorithms compared with the ABC is calculated by Equation (16). In Figure 11, it can be clearly seen that PSO and CS perform worse than ABC, while the performance of GA, ABC-AI and ABC-TS is very close, slightly better than ABC, but significantly worse than HABC. In other words, the proposed algorithm achieves the best results among all of the algorithms in all instances, which proves again that the proposed algorithm is suitable for this specific problem.

From the two experiments, it is safe to conclude that the proposed algorithm can achieve a gap of less than 5% from the exact solution for small-size instances, which is acceptable in reality. Besides, the proposed HABC algorithm performs better than other metaheuristic algorithms, achieving about 5–30% improvement in different instances owing to the improved procedure described in Section 4.

6. Conclusions and future works

This study deals with cooperative task planning of heterogeneous multi-robot systems in an order picking warehouse composed of a fleet of autonomous robots and a group of workers, and formulates the problem as an open shop scheduling with sequence-dependent set-up and transportation times, with the objective of minimizing the makespan. A MILP model is established to obtain the optimal scheme of cooperative task planning of heterogeneous MRSs. To increase the size of solvable problems and improve the performance of the solution, an effective HABC is developed. Through comparative experiments, it is verified that these three metaheuristic algorithms (ABC-AI, ABC-TS and HABC)

achieve significant improvements compared with the ABC algorithm. Besides, as the problem size increases, the proposed HABC algorithm gradually exceeds the MILP solution in performance in the same amount of time, and achieves better results than other metaheuristic algorithms in almost all instances. This further demonstrates the effectiveness of the proposed algorithm.

This study makes an important contribution to the overall knowledge in the field of cooperative task planning of heterogeneous multi-robot systems. From another point of view, the development of the HABC also expands the application scope of the ABC algorithm. Although this research has dealt with several academically challenging issues, future work is still required. The first extension is to consider more complicated scenarios in an order picking warehouse, such as multiple people in each workshop, and some other extensions based on this assumption. Furthermore, owing to the influence of dynamic interruptions in actual task execution, real-time planning or replanning based on the up-to-date shop floor information will be one of the directions for future research.

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